

Chapter 2: Data and Data Pretreatment

Outline

Introduction

What is Data?

Process Data

Modeling Process Data

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Data Pretreatment

2.1 Introduction

- ▶ Data mining and Machine Learning analyze large amounts of multivariate process data collected in modern industrial databases.
- ▶ Often the raw data are not ready for algorithmic use and must be preprocessed.
- ▶ Typical preprocessing tasks include outlier removal, missing-data approximation, and normalization to avoid unit-driven bias.
- ▶ Sensor and sampling errors should be addressed and preprocessing automated for reproducibility and efficiency.

2.1 Introduction (continued)

- ▶ Batch-process data are time-varying and require methods different from near-steady-state techniques.
- ▶ Fault-detection datasets often present severe class imbalance (many normal examples, few faults), which needs specialized sampling and learning strategies.
- ▶ This chapter reviews outlier removal, imputation, sampling, and scaling techniques relevant to process data.

2.2 What is Data?

- ▶ Data is a collection of data objects described by a number of attributes.
- ▶ Attributes (features, variables), are properties or characteristics of an object that may vary from one object to another as well as over time.
- ▶ Each element in the collection is also called a record or data point. Each record is a vector of attribute values.
- ▶ So, data is the set $\mathcal{D} = \{a_{1,1}, \dots, a_{n,m}\}$, with n records and m attributes.
- ▶ In modeling some attributes are inputs and others are outputs depending on the problem. Selection of inputs/outputs depends on causal knowledge and domain expertise.

2.2 What is Data? (scales and units)

- ▶ Attributes use measurement scales (e.g., Kelvin, Celsius) and have different magnitudes (e.g., temperature, mass).
- ▶ Without scaling, large-magnitude variables can dominate analysis.
- ▶ Scaling/normalization methods are used to make variables comparable.

2.3 Process Data — instrumentation

- ▶ Plant instruments are the engineer's “eyes”; their data enable process understanding and monitoring.
- ▶ Process measurements commonly appear as time series (run charts) used to inspect centerline, variability, trends, and transients.
- ▶ Consider measurement error, accuracy (bias), and precision when assessing data quality.

2.3 Process Data — distributions

- ▶ Use histograms or probability density functions to study data distribution and concentration.
- ▶ Distributional shape (skewness, multimodality) informs model selection and preprocessing decisions.
- ▶ Comparing process histograms with reference distributions (e.g., normal) helps identify departures from assumptions.

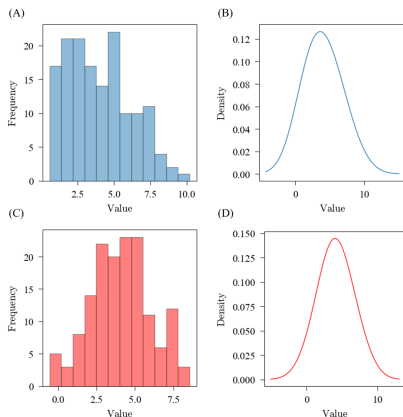


Figure: Example of a process data histogram.

2.4 Modeling Process Data

- ▶ Conceptual model:
observation = mean (actual) level + random deviation
- ▶ Typical assumptions: zero-mean errors, constant variance, independence; Gaussianity often assumed but should be checked.

$$y = \bar{y} + \varepsilon(i), i = 1, 2, \dots \quad (1)$$

$$\varepsilon(i) \sim \mathcal{N}(0, \sigma^2) \quad (2)$$

2.4 Modeling Process Data (implications)

- ▶ If deviations are Gaussian, mean and variance summarize the signal.
- ▶ The 1/2/3-sigma rules (68/95/99.7) underpin control-charting thresholds.
- ▶ For multivariate signals, specialized multivariate monitoring methods are required.

2.5 Issues with Plant Data

- ▶ Raw plant data can have missing points, gross errors, and outliers from instrumentation, sampling, or transmission faults.
- ▶ Common issues include drift, noise, multicollinearity, and extended blocks of missing values.
- ▶ Such issues should be detected and treated prior to modeling.

2.6 Data Pretreatment — Overview

- ▶ Key pretreatment operations:
 - ▶ Missing-data approximation
 - ▶ Outlier handling
 - ▶ Sampling (under/oversampling)
 - ▶ Normalization/standardization
- ▶ Choice and order depend on dataset size, imbalance level, and downstream modeling needs.

2.6.1 Missing-data Approximation

- ▶ For large datasets, missing values may be ignored; for smaller datasets they often need imputation.
- ▶ Simple methods: mean/median imputation, forward/backward fill.
- ▶ Advanced methods: regression-based imputation and multiple imputation.
- ▶ Interpolating large contiguous gaps can misrepresent the system. Consider downstream effects when choosing an imputation strategy.

2.6.2 Outlier Detection and Removal

- ▶ Outliers are observations that do not follow the bulk distribution and can strongly influence estimates.
- ▶ Treatment options: remove, clip, or model outliers separately.
- ▶ Detection methods include Z-score, robust estimators, statistical tests, T2, and clustering-based approaches.

$$z_i = \frac{x_i - \bar{x}}{\sigma} \quad (3)$$

2.6.2 (cont.) — k-NN and elbow method

- ▶ k-NN based detection flags points that are far from their neighbors.
- ▶ The elbow method on sorted k-NN distances can be used to choose a threshold for outlier identification.
- ▶ The choice of k affects sensitivity and must be validated on data.

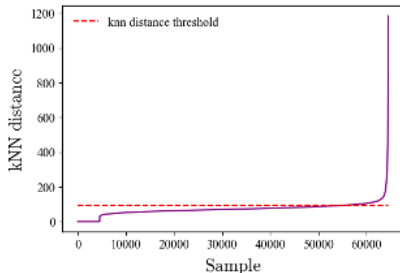


Figure: Elbow method on sorted k-NN distances.

2.6.3 Under-sampling and Oversampling

- ▶ Class imbalance can severely bias classifiers when faults are rare.
- ▶ Under-sampling reduces majority-class instances; oversampling increases minority-class samples.
- ▶ Sampling methods include random sampling, interval averaging, and condensed nearest neighbors (c-NN) sampling to preserve important dynamics.
- ▶ Synthetic minority oversampling technique (SMOTE) generates minority samples between nearest neighbors to enlarge minority-class decision regions.

2.6.4 Standardization and Normalization

- ▶ Scaling places variables on comparable scales to avoid magnitude dominance.
- ▶ Standardization (z-score) centers and rescales to unit variance; useful when distributions are approximately Gaussian.
- ▶ Normalization (min-max or divide-by-max) rescales values into a bounded range without distributional assumptions.

2.7 Conclusions

- ▶ Data preprocessing is a critical step for reliable ML in process industries.
- ▶ Sampling, outlier treatment, imputation, and scaling all influence model performance and should be chosen carefully.
- ▶ Align preprocessing choices with downstream tasks and data characteristics.